Developing Domain-Specific NER Models for the British Alpine Corpus: Annotation, Training, and Evaluation Using Prodigy

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Abstract This study developed a domain-adapted named entity recognition system for the British Alpine Corpus using the Prodigy and SpaCy frameworks, achieving an F1 score of 90.85% on 1,060 sentences of training data. This result represents a substantial improvement over existing general-purpose NER models evaluated on the same corpus. The system provides high-quality language resources for both historical and contemporary mountaineering literature research.

Keywords: Named Entity Recognition, British Alpine Corpus, Prodigy, SpaCy

1. Introduction

In natural language processing, named entity recognition (NER) is a fundamental task that makes it possible to automatically extract and classify entities such as people, places, and organizations. Modern NER algorithms perform well on texts in the general domain, but they become significantly less successful when applied to specialized areas, such as historical climbing literature. As Ehrmann et al. (2021) point out, historical documents pose unique challenges to NER systems due to their linguistic diversity and noisy inputs.

By creating a domain-adapted NER system for the British Alpine Corpus (BAC) using the Prodigy annotation platform and SpaCy framework, this research aims to address this gap and provide linguistic resources of superior quality that can be reused for future research. The study contributes to the broader goal of adapting NLP techniques to specialized historical and technical domains.

2. Background

The British Alpine Corpus (BAC) is a digital archive comprising expedition reports and Alpine Journal entries from the late 19th to early-21st century, documenting British mountaineering expeditions in the Alps and other major mountain ranges. This corpus holds significant historical and linguistic value as it captures the evolution of mountaineering terminology and practices during the golden age of alpinism.

However, the BAC presents unique challenges for NER systems. As digitized historical documents, the texts contain numerous optical character recognition (OCR) artifacts including scanning errors, irregular spacing, and fragmented words. OCR noise and historical digitisation artefacts are common in the source material. All of them require extensive preprocessing. The domain-specific nature of the content introduces complex entity recognition scenarios: toponyms such as “Sciora” refer to mountain groups rather than persons, while Welsh compound names such as “C’logwyn Du’r Arddu” demand specialized labeling. Expedition-specific terminology such as “132 man-days” represents temporal concepts that conventional NER schemas often fail to capture.

Existing NER methods show significant limitations when analyzing such material, as general-purpose models trained on news or web data lack the domain-specific knowledge required. This research focuses on methods tailored to Alpine historical texts, expanding current approaches in domain adaptation for NER. It leverages active learning via the Prodigy AI platform, combining corpus linguistics with contemporary machine learning techniques.

3. Experiments

This section introduces the proposed annotation scheme and documents the experiments conducted to train NER models on gold-standard corpora of varying sizes and time periods.

Table 1: Annotation Scheme for Alpine Corpus

Entity Type	Definition
PERSON	A named individual human being, including climbers and historical figures (e.g., Edward Whymper).
MOUNTAIN	A named natural elevation of the earth's surface, such as peaks or summits (e.g., Matterhorn). Named features of the mountain landscape used by climbers and alpinists as locations for ascents and share naming patterns with mountain features (e.g., North Face, Eigerwand).
VALLEY	A named low area between hills or mountains, often with a river running through it (e.g., Lauterbrunnental).
CITY	A named human settlement size and administrative structure (e.g., Geneva).
GPE	A country, state, province, or region with a political or geographic boundary (e.g., Switzerland, Canton of Bern).
DATE	A specific calendar-based reference that identifies a day, a month, year, or a combination thereof, used to pinpoint an event or mention in time (e.g., 14 July 1865, in 1923, March 1897, the summer of 1880). Pay special attention to the difference between DURATION and TIME.

3.1 Models Trained on a Small Gold Standard (415 & 867 sentences)

The experiments test the robustness of the annotation scheme using limited data. To assess the scheme's temporal adaptability, I compare models trained on historical texts (1969-2008) and modern texts (2020-2022).

3.2 Models Trained on a Medium Gold Standard (1,060 sentences)

These experiments validate the scheme's scalability using the same temporal splits to evaluate how increased data volume improves model recognition of domain-specific named entities across both historical and contemporary texts.

3.3 Models Trained on a Large Gold Standard (2,300 sentences)

The large-scale experiments further examine the scheme's generalizability and performance at scale. We assess whether expanded training data enhances model consistency in entity recognition across contemporary texts.

4. Results and Discussion

All five models trained on different gold standard sizes and data periods are summarized in Table 2. Models_01 and Model_02 were trained on the SpaCy baseline model for English (en_core_web_sm). Model_03 was trained on Model_01/model-best. Models_04 and Model_05 were trained on Model_02/model-best. For each model, the result with the highest F1-score across all epochs is reported, as this metric best reflects the model's ability to balance precision and recall.

Table 2: Performance of All Trained Models

Model	Gold Standard Size / sentence	Text Period	Precision	Recall	F1 Score
Model 1	415	1969-2008	81.33%	82.43%	81.88%
Model 2	867	2020-2022	80.53%	81.98%	81.25%
Model 3	1,060	1969-2008	85.92%	81.88%	83.85%
Model 4	1,060	2020-2022	90.85%	90.85%	90.85%
Model 5	2,301	2020-2022	82.09%	82.09%	82.09%

4.1 Effects of Training Data Scale

The results reveal a non-monotonic relationship between training data volume and model performance. Model 4, trained on 1,060 sentences of contemporary texts, achieved the highest F1-score of 90.85%. In contrast, Model 5, trained on 2,301 sentences of contemporary texts, achieved 82.09%. This pattern indicates that increasing data volume alone does not guarantee improved performance; other factors such as data quality, annotation consistency, and the temporal distribution of the corpus appear to play important roles. The unexpectedly lower performance of Model 5 may also reflect increased annotation heterogeneity introduced by the larger dataset, particularly as OCR artefacts and boundary decisions become more variable across a broader range of texts. One possible explanation for this non-monotonic trend is that variance in precision values may increase with smaller datasets. Given the relatively limited size of the dataset used in this study, individual errors or anomalies during annotation can have a more pronounced effect on overall metrics. However, the consistent ranking of Model 4 as the highest-performing model across all experiments suggests that factors beyond data volume are at work. Further investigation would be needed to fully understand this pattern, possibly through detailed error analysis and examination of annotation consistency across the different datasets.

4.2 Effects of Text Period

Models trained on contemporary texts (2020–2022) generally achieved superior performance compared to those trained on historical texts (1969–2008). Model 2 (F1 = 81.25%) and Model 4 (F1 = 90.85%) both achieved higher scores than their historically-trained counterparts, though Model 3 trained on 1,060 sentences of historical texts achieved an F1-score of 83.85%, demonstrating that well-annotated historical material can still yield reasonably strong performance. The performance gap likely reflects differences in text quality, OCR preprocessing, or consistency in terminology usage across the two periods; contemporary texts being more recently produced may have been more carefully transcribed or contain fewer OCR artifacts than their historical counterparts.

Upon reflection, the temporal division may have been less important than initially anticipated for the purposes of NER model development. The gap between 2008 and 2020 does not represent a fundamental shift in how the Alpine Club Journal documents mountaineering activities or in the linguistic patterns present in such texts. This temporal categorization was implemented to explore whether the annotation scheme could adapt across different periods. However, given that the ultimate objective was to develop a high-quality domain-specific NER model for the Alpine corpus as a whole, a simpler unified approach might have been equally valid and potentially more efficient.

4.3 Comparison with Prior Work

Previous work by Ay (2021) evaluated SpaCy NER models, both statistical and transformer-based, on the British Alpine Corpus. The transformer model achieved an F1-score of 0.77 under strict matching conditions and 0.88 under type-based matching. The statistical model performed notably worse, with F1-scores of 0.65 (strict) and 0.75 (type-based).

The current study's best model (Model 4) achieved an F1-score of 90.85%, substantially exceeding both the transformer and statistical models evaluated in prior work. This improvement likely results from a combination of domain-specific annotation scheme design, careful manual annotation, and selection of contemporary texts for training. The custom annotation categories (MOUNTAIN, VALLEY, DATE) were specifically tailored to capture entities relevant to mountaineering, whereas prior approaches relied on general-purpose NER schemas.

These findings suggest that modest investments in domain-specific annotation and careful corpus selection can yield substantial improvements in NER performance for specialized domains, even when working with limited training data.

5. Conclusion

This study successfully developed a domain-adapted NER system for the British Alpine Corpus, achieving an F1-score of 90.85% and demonstrating substantial improvements over previously published results on the same corpus. The resulting annotation datasets were exported both as JSONL files for model development and as TEI-compatible XML files for corpus preservation and digital humanities applications. The research highlights several key findings.

First, domain-specific annotation schemes tailored to the characteristics of specialized corpora can substantially improve NER performance. The custom entity types (MOUNTAIN, VALLEY, DATE) captured nuances in mountaineering terminology that general-purpose schemas could not adequately represent.

Second, the relationship between training data volume and model performance is not strictly monotonic. While sufficient data is necessary for model training, other factors such as text period, data quality, and corpus composition also play critical roles.

Third, annotation challenges persist even with careful schema design. Multi-word entities such as Sciora hut or Gianetti hut path present ongoing challenges. Deciding whether to label Sciora as an individual named entity or to consider Sciora hut as a complete entity requires human judgment and domain expertise. These decisions must ultimately be resolved on a case-by-case basis, guided by well-documented annotation guidelines.

When developing or applying NER models to specialized domains, careful selection of training corpora and appropriate model architecture selection can substantially improve entity recognition performance. Future research may benefit from exploring domain-specific optimization techniques and investigating the underlying factors that contribute to the non-monotonic relationship observed between training data volume and performance.

6. Ethical Considerations and Limitations

6.1 Annotation Scheme Coverage

In this research, the number of named entity categories was deliberately limited to six to achieve an optimal balance between granularity and annotation consistency. Although the categories GPE (Geopolitical Entity) and CITY were clearly defined and distinguishable in most contexts, the annotation process revealed that additional entity types warrant consideration for Alpine texts. Specifically, VILLAGE and TOWN represent distinct categories that occur frequently in mountaineering literature and could be distinguished from broader geopolitical entities. Furthermore, GLACIER emerged as a significant entity type during annotation, yet was not formally included in the final scheme. For future annotation efforts, incorporating VILLAGE, GLACIER, and other specialized entities would better capture the specific characteristics of Alpine expeditions and journals, thereby enhancing the granularity and accuracy of NER models tailored to this domain.

6.2 Data Collection and Annotation Effort

The author initially aimed to create an 8,000-sentence gold-standard dataset but had to scale down this target as the deadline approached. This adjustment underscores the practical constraints inherent in academic research and highlights the broader disparity between research settings and industrial NER practices. In industry, annotation tasks are typically supported by larger, well-funded teams and extensive resources. The field of Natural Language Processing continues to suffer from a severe shortage of labeled data due to the highly expensive and time-consuming nature of manual annotation. This observation is equally relevant to NER, where resource limitations often necessitate compromises in both data volume and annotation depth.

Human annotation, especially by a single annotator, has inherent limitations. From a quantitative research perspective, achieving a gold-standard dataset of over 1,000 sentences is highly demanding, particularly when manual annotation is both time-consuming and cognitively demanding. Moreover, it is unrealistic to exhaustively list all relevant terminology and examples unless one is a dedicated expert in Alpine corpus research with years of effort devoted to building a specialized lexicon. Ethically speaking, modern NLP techniques are not intended to replace such expert roles; however, avoiding collaboration with AI tools is equally unrealistic. The path forward involves leveraging artificial intelligence in tandem with human expertise and domain knowledge.

6.3 Annotation Scheme Formalization

This study employed a custom annotation scheme developed specifically for the British Alpine Corpus rather than adopting a formally established guideline such as CoNLL or

OntoNotes. While the scheme proved effective for the British Alpine Corpus, further validation across additional Alpine and mountaineering corpora would be required before it could be considered a broadly applicable domain-specific annotation standard. The scheme's validity is demonstrated through successful model training and competitive performance metrics. However, for broader adoption and comparison with other research, future work should consider formal documentation of the scheme, publication of annotation guidelines, and potentially inter-annotator agreement studies to establish the scheme's reliability and generalizability.

7. Acknowledgements

I would like to sincerely thank my supervisor Martin for his valuable insights on comparative research methodology and his preliminary guidance on named entity selection. I am also grateful for the support throughout this research journey, including the development of annotation guidelines and model evaluation protocols.

8. Appendix

8.1 Link to the Manually annotated BAC NER datasets

https://github.com/barcetonio6668/Annotation-Training-and-Evaluation-Using-Prodigy/tree/main/Checked_Annotations

8.2 Links to Trained SpaCy NER

Models for Future Use

Model_01 is located at:
https://github.com/barcetonio6668/Annotation-Training-and-Evaluation-Using-Prodigy/tree/main/model_01

Model_02 is located at:
https://github.com/barcetonio6668/Annotation-Training-and-Evaluation-Using-Prodigy/tree/main/model_02

Model_03 is located at:
https://github.com/barcetonio6668/Annotation-Training-and-Evaluation-Using-Prodigy/tree/main/model_03

Model_04 is located at:
https://github.com/barcetonio6668/Annotation-Training-and-Evaluation-Using-Prodigy/tree/main/model_04

Model_05 is located at:
https://github.com/barcetonio6668/Annotation-Training-and-Evaluation-Using-Prodigy/tree/main/model_05

8.3 Links to the Documentation of Prodigy Pipeline and the Annotation Guidelines

README.md is located at:
https://github.com/barcetonio6668/Annotation-Training-and-Evaluation-Using-Prodigy/blob/main/README.md

Annotation Guidelines in .docx are located at:
https://github.com/barcetonio6668/Annotation-Training-and-Evaluation-Using-Prodigy/tree/main/Annotation%20Guidelines

9. References

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